

Problems

3-1: Obtain the simplified expressions in sum of products for the following Boolean functions:

(a) $F(x, y, z) = \Sigma(2, 3, 6, 7)$
 $= y$

yz	00	01	11	10
0			1	1
1			1	1

(b) $F(A, B, C, D) = \Sigma(7, 13, 14, 15)$
 $= BCD + ABD + ABC$

CD	00	01	11	10
00				
01			1	
11		1	1	1
10		1	1	1

(c) $F(A, B, C, D) = \Sigma(4, 6, 7, 13)$
 $= BCD + A'BD'$

CD	00	01	11	10
00				
01	1		1	1
11			1	
10				

(d) $F(w, x, y, z) = \Sigma(2, 3, 12, 13, 14, 15)$
 $= w'xz + wz$

xz	00	01	11	10
00			1	1
01				
11	1	1	1	1
10				

3-2: Obtain the simplified expressions in sum of products for the following Boolean functions:

a) $xy + x'y'z' + x'yz' = xy \cdot 1 + x'y'z' + x'yz'$ (Postulate 2(b))
 $= xy \cdot (z + z') + x'y'z' + x'yz'$ (Postulate 5(a))
 $= xyz + xy'z' + x'y'z' + x'yz'$
 Simplified: $xy + x'z'$

yz	00	01	11	10
0	1			1
1			1	1